

CASPER FARRET JENTINK

Postdoctoral Researcher, Department of Astronomy, University of Geneva

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RESEARCH INTERESTS

Exoplanet atmospheric escape; the metastable helium triplet at 1083 nm; high-resolution near-infrared spectroscopy; astronomical instrumentation and optical design.

APPOINTMENTS

2025–present **Postdoctoral Researcher**, Department of Astronomy, University of Geneva, Switzerland
Project lead of the NIGHT spectrograph.

EDUCATION

2021–2025 **PhD in Astronomy & Astrophysics**, University of Geneva, Switzerland
Thesis: *The Design and Development of NIGHT: A High-Resolution Near-Infrared Spectrograph for Exoplanetary Atmospheric Escape Studies via Helium Triplet Spectroscopy*
Advisors: Prof. Francesco Pepe & Prof. Christophe Lovis
Defended 12 November 2025 with distinction *très bien* (highest)
Awarded the **Edith Alice Müller Award 2026** (SSAA) for an outstanding PhD thesis

2018–2021 **MSc in Astronomy**, Kapteyn Astronomical Institute, University of Groningen, The Netherlands
Thesis: development of ABORAS, a polarimetric Sun-as-a-star telescope for HARPS3
Supervisors: Prof. Frans Snik & Dr. Willem Jellema

2015–2018 **BSc in Astronomy**, Kapteyn Astronomical Institute, University of Groningen, The Netherlands
Thesis: construction of the backend of the Kapteyn Interferometer for Short-baseline Solar observations

AWARDS & HONOURS

2026 **Edith Alice Müller Award**, Swiss Society for Astrophysics and Astronomy (SSAA), for an outstanding PhD thesis in astronomy in Switzerland

2023 **Award for the EXPOSURE mission concept**, Alpbach Summer School

2021 **Beta Business Days Tech Start-up Award**

2021 **Kapteyn Science Day Award**

GRANTS

2026	MERAC Funding and Travel Award	CHF 2,000
2024	SPIE Travel Grant	USD 1,000
2023	NCCR PlanetS Open Call Grant For the development of NIGHT; PI; success rate 15 %	CHF 135,000

TELESCOPE TIME & OBSERVING EXPERIENCE

2025–2026	54 nights on the 1.52 m telescope, Observatoire de Haute-Provence (France), secured for NIGHT commissioning and science operations (lead)
2021–2025	40+ nights as observer at the Swiss 1.2 m Euler telescope, La Silla Observatory (Chile)

INSTRUMENTATION PROJECTS & RESEARCH EXPERIENCE

2021–present	Project lead, NIGHT (Near-Infrared Gatherer of Helium Transits), University of Geneva Designed and built a compact, fiber-fed, high-resolution near-infrared spectrograph dedicated to surveying the metastable helium triplet in exoplanet atmospheres. Led an international team of scientists and engineers; collaborated with industrial partners on novel instrumentation for fiber-fed high-resolution spectroscopy, securing >CHF 100,000 in in-kind hardware contributions. First light at the OHP 1.52 m telescope in 2025. Negotiated the installation of NIGHT at the 3.58 m Telescopio Nazionale Galileo (TNG), La Palma.
2025–2026	NUV spectrograph assessment, WALTzER , ESA F-class mission concept Preliminary assessment of the near-ultraviolet spectrograph: optical design of a toroidal concave-grating spectrograph covering 240–325 nm at $R > 2800$, with trade studies of the shutter mechanism, dichroic and cross-dispersion sampling.
2021	Project lead, ABORAS , polarimetric Sun-as-a-star telescope for HARPS3
Feb.–Jul. 2020	Research intern , SRON Netherlands Institute for Space Research & TU Delft Micromachined dielectrics for SPICA/SAFARI: simulation and evaluation of an effective dielectric material for a resonant immersed waveplate.
2017–2021	Observatory assistant , Blaauw Observatory, University of Groningen

PUBLICATIONS

	6 refereed papers (3 first-author), 6 conference proceedings (3 first-author). <i>Refereed – first author</i>
2026	Farret Jentink, C. , Bourrier, V., & Carteret, Y., <i>Probing Atmospheric Escape Through the Near-Infrared Helium Triplet</i> , arXiv:2602.03667
2025	Farret Jentink, C. , Pepe, F., Lovis, C., Schwab, C., Wildi, F., & Clawson, A., <i>High-resolution, high-efficiency narrowband spectroscopy with an s-p-phased holographic grating in double pass</i> , <i>A&A</i> , 698, A249
2024	Farret Jentink, C. , Bourrier, V., Lovis, C., Allart, R., Chazelas, B., Lendl, M., Dumusque, X., & Pepe, F., <i>NIGHT: A compact, near-infrared, high-resolution spectrograph to survey helium in exoplanet systems</i> , <i>MNRAS</i> , 527, 4467

Refereed – co-author

- 2025 Frensch, Y. G. C., Bouchy, F., Lo Curto, G., et al. (incl. **Farret Jentink, C.**), *Three hot Jupiters transiting K-dwarfs with significant heavy element masses*, *A&A*, 700, A118
- 2025 Fontanet, E., Udry, S., Ségransan, D., et al. (incl. **Farret Jentink, C.**), *CORALIE radial-velocity search for companions around evolved stars (CASCADES). IV. New planetary systems around HD 87816, HD 94890, and HD 102888 and an update on HD 121056*, *A&A*, 699, A38
- 2025 Seidel, J. V., Prinoth, B., Pino, L., et al. (incl. **Farret Jentink, C.**), *Vertical structure of an exoplanet's atmospheric jet stream*, *Nature*, 639, 902

Conference proceedings – first author

- 2026 **Farret Jentink, C.**, Pepe, F., Lovis, C., et al., *NIGHT at the OHP152 Telescope: a dedicated high-resolution near-infrared spectrograph for exoplanetary atmospheric escape*, *Proc. SPIE*, 14149, 1414906
- 2024 **Farret Jentink, C.**, Pepe, F., Lovis, C., et al., *The Near-Infrared Gatherer of Helium Transits (NIGHT)*, *Proc. SPIE*, 13096, 1309670
- 2022 **Farret Jentink, C.**, Mortier, A., Snik, F., Dorval, P., Thompson, S. J., Navarro, R., & Naylor, T., *ABORAS: polarimetric, 10 cm/s RV observations of the Sun as a star*, *Proc. SPIE*, 12182, 1218231

Conference proceedings – co-author

- 2024 Hagelberg, J., Restori, N., Ségransan, D., et al. (incl. **Farret Jentink, C.**), *On-sky performance of KalAO, the swift adaptive optics imager on the 1.2 m Euler Swiss telescope in La Silla, Chile*, *Proc. SPIE*, 13097, 130970C
- 2024 Dorval, P., Politopoulos, I., Nicholson, B., ter Horst, R., **Farret Jentink, C.**, et al., *The polarimetric performance of HARPS3 and its impact on the Terra Hunting Experiment scheduling*, *Proc. SPIE*, 13096, 130968C
- 2022 Jellema, W., Ridder, M., Eggens, M., **Farret Jentink, C.**, et al., *A Broadband Resonant Half-Waveplate for Compact Far-Infrared Grating Spectrometers with Continuous Wavelength Band Division*, *47th Int. Conf. on Infrared, Millimeter and Terahertz Waves (IRMMW-THz)*, 582

SOFTWARE

NIGHTMARE: instrument control software for NIGHT

NIGHTCAP: data reduction pipeline for NIGHT

TALKS (SELECTION)

- Sep. 2026 Contributed talk, CHArting the future of Stellar and Exoplanet Spectroscopy (CHASES2026), ESO/INAF/OCA conference, Elba, Italy
First light of the NIGHT spectrograph at the 152cm telescope of OHP
- Jul. 2026 Contributed talk, SPIE Astronomical Telescopes + Instrumentation 2026, Copenhagen, Denmark
NIGHT at the OHP152 Telescope: a dedicated high-resolution near-infrared spectrograph for exoplanetary atmospheric escape
- Jun. 2026 Invited talk, ARIEL Science Day France, Institut d'Astrophysique de Paris (IAP), Paris, France
The NIGHT Project
- Apr. 2025 Invited talk, NCCR PlanetS General Assembly, Neuchâtel, Switzerland
- Apr. 2024 Invited talk, NCCR PlanetS General Assembly, Engelberg, Switzerland

CONFERENCES & SCHOOLS ATTENDED (SELECTION)

- Sep. 2026 CHASES2026 (ESO/INAF/OCA conference), Elba, Italy
- Jul. 2026 SPIE Astronomical Telescopes + Instrumentation, Copenhagen, Denmark
- Jun. 2024 SPIE Astronomical Telescopes + Instrumentation, Yokohama, Japan
- Sep. 2023 Spectral Fidelity, Florence, Italy
- 2023 Alpbach Summer School, Alpbach, Austria
- Apr. 2022 Extreme Precision Radial Velocities V (EPRV V), Santa Barbara, USA

TEACHING

- 2022–2026 **Teaching assistant**, Observational Techniques (MSc), University of Geneva
Course by Prof. Francesco Pepe, covering spectroscopy, adaptive optics, light detection, interferometry and polarimetry.
- 2023 **Teaching assistant**, Observations, Data Acquisition (MSc) & Analysis, University of Geneva
Course by Prof. Laurent Eyer, covering the data processing of the GAIA catalogue.
- 2020–2021 **Teaching assistant**, The Evolving Universe (BSc), University of Groningen
Course by Prof. Marc Verheijen, covering a summary of modern astronomy for the astronomy minor.
- 2018 **Astrophysics teacher**, Boswell Beta, Utrecht, The Netherlands

STUDENT SUPERVISION & MENTORING

- 2026–present **Angelos Mengoudis**, MSc thesis, University of Geneva
Co-supervised with Prof. Christophe Lovis and Prof. Frans Snik.
- 2026 **Angelos Mengoudis**, MSc astrophysics laboratory (APL) project, University of Geneva
- 2023 **Mariam Haidar**, MSc thesis, University of Geneva
Guidance during the MSc thesis (informal; not official supervisor).
- 2023 **Antonin Besse**, MSc astrophysics laboratory (APL) project, University of Geneva

2019–2020 **First-year student mentor**, University of Groningen

PUBLIC OUTREACH & MEDIA

Jun. 2026 Invited public talk to celebrate Pride and LGBTQIA+ in Astronomy, Astronomy on Tap, Bern, Switzerland: *Reading the Rainbow*

Sep. 2025 Documentary, VPRO Tegenlicht (Dutch national broadcaster)

Sep. 2025 University of Geneva press release on NIGHT first light

Apr. 2025 Interview, NRC (Dutch national newspaper)

Sep. 2024 Invited public talk, Scientifika, Stockholm, Sweden

2018–2021 Operator, DOT Liveplanetarium, University of Groningen

2017–2021 Operator, Kapteyn Mobile Planetarium, University of Groningen

SERVICE & LEADERSHIP

2019–2021 **Board member**, JWG, Dutch youth association for astronomy

2017–2020 **Sirius A**, study association of astronomy, University of Groningen
Vice-chair & commissioner of internal affairs (2017–2018); chair of the board (2018–2019; budget ~EUR 20k, >250 members); advisory board (2019–2020).

2021 **Local policy coordinator**, education & culture, Volt Netherlands

SKILLS

Instrumentation Optical design, (fiber-fed) (high-resolution) spectroscopy, (VPH) gratings, near-infrared detectors, polarimetry, optics/lithography

Software Python, MATLAB, $\text{\LaTeX}$, Zemax, COMSOL, Adobe CC

Languages Dutch (native), English (fluent), German (proficient), French (intermediate)

REFERENCES

Available on request.

Last updated: September 2026